

ANSHAY AGARWAL

Senior AI Systems Engineer | Robotics, Sensors, Medical AI | Ex-Nvidia/Qualcomm | [LinkedIn](#) | [GitHub](#)
anshayagr@gmail.com | +91-8800472674

SUMMARY

R&D Scientist and Systems Engineer with 10+ years of experience (Ex-Qualcomm, Nvidia, DRDO) in **Computer Vision, Signal Processing, and Real-Time Systems**. Deep expertise in **Multi-Modal Sensor Fusion** (X-Ray/RGB/Thermal) and **Deep Learning** for detection. Passionate researcher in **Medical Imaging (CT/MRI)** and **Neurotechnology (BCI)** with proven success in deploying high-performance algorithms on edge hardware.

MEDICAL AI RESEARCH

Vitalis: Edge-AI Diagnostic Platform | Lead Developer

- Engineered a **multi-modal**, on-device mobile application deploying **Google's HeAR and MedSigLip models** for early stage detection of diseases using smartphone sensors.
- Designed complex **sensor-fusion** pipelines to extract physiological biomarkers directly from raw smartphone microphone and camera arrays for immediate clinical triaging.

Automated Arterial Segmentation (CT Scans) | Deep Learning Researcher

- Developed a **3D Semantic Segmentation pipeline** using **PyTorch** to isolate Carotid and Vertebral arteries from volumetric CT Angiography data.
- Implemented **Calcification Classification algorithms**, utilizing deep neural networks to quantify arterial plaque burden, aiding in automated stroke risk assessment.

Brain-Computer Interface (BCI) for Motor Control | Researcher

- Designed a real-time **Motor Imagery BCI system** acquiring non-invasive EEG signals to decode user intent.
- Implemented Signal Processing pipelines to extract features from raw EEG data, enabling direct thought-controlled cursor manipulation in a virtual environment.

EXPERIENCE

Greyscale AI | Senior Computer Vision Engineer | Feb 2021 - Oct 2022

- Multi-Modal Sensor Fusion:** Architected the software stack to synchronize and merge data from **X-Ray and RGB sensors**, enabling automated foreign object detection in real-time streams.
- Edge AI Deployment:** Developed the complete embedded software pipeline handling real-time communication between hardware sensors and the inference engine.

Qualcomm | Senior Lead Software Engineer | Apr 2023 - Jan 2025

- Optimized Real-Time Rendering:** Achieved a **96% reduction in latency (35ms → 1.3ms)** for XR Flicker correction on GPU, critical for high-fidelity visual systems.
- Advanced Imaging Algorithms:** Led the design of novel HDR flows for Premium Tier chips, optimizing signal processing chains for next-gen hardware.

Schoolbook.ai | Co-Founder, Head of AI and Engineering | AI, Python | Jun 2025 - Sep 2025

- **RAG & GenAI Architecture:** Built a multimodal RAG pipeline integrating Vector Databases and Gemini Pro to synthesize technical content from unstructured datasets.
- **Agentic Workflows:** Engineered autonomous AI agents using **LangGraph** capable of multi-step reasoning and tool use

NVIDIA | System Software Engineer | Jul 2017 - Jun 2018

- **Imaging Stack Optimization:** Enhanced the Tegra Camera imaging stack, resolving critical **Lens Shading** and **Bayer Demosaicing** artifacts for high-precision computer vision applications

DRDO | Scientist | Aug 2013 - Jul 2015

- **Thermal Imaging R&D:** Engineered image enhancement algorithms for **Night Vision Thermal Cameras**, improving visibility range up to 40km.
- **Signal Restoration:** Developed algorithms to correct Vignette artifacts and improve Signal-to-Noise Ratio (SNR) in low-contrast thermal data.

SKILLS

Core: Computer Vision, Signal Processing, Deep Learning, Real-Time Systems.

AI/ML: PyTorch, TensorFlow, OpenCV, LangChain, RAG, Transformers.

Medical/Bio: Volumetric Segmentation (3D U-Net), EEG Signal Processing.

Languages: Python, C++.

EDUCATION

- **IIT Delhi: M.Tech, Computer Technology (2015 - 2017)**
- **IIT Mandi: B.Tech, Electrical Engineering (2009 - 2013)**

SPECIALIZATIONS

- **Wharton School:** Entrepreneurship Specialization (2021-2022)
- **DeepLearning.AI:** TensorFlow Professional Cert (2020), Deep Learning Specialization (2019)
- **Johns Hopkins University:** Fundamental Neuroscience for Neuroimaging.
- **Hebrew University of Jerusalem:** Synapses, Neurons and Brains.
- **Northwestern University:** Modern Robotics (Motion Planning & Control).

ACHIEVEMENTS

- **Top Ranks:** GATE (AIR 423), IIT-JEE (Top 1%), GRE (321/340), XII Board (Top 0.1% Math).